

Ontology Profile Validation, Reasoning, and Namespace Analysis with ROBOT

The command examples in this report use these shell variables:

```
export REPO_ROOT="$(pwd)"
export ROBOT="$REPO_ROOT/bin/robot"
export FASTO="$REPO_ROOT/data/FASTO_RDF.owl"
export QUERY_DIR="$REPO_ROOT/queries"
export RESULTS_DIR="$REPO_ROOT/results"
mkdir -p "$RESULTS_DIR"
```

Validation and reasoning procedures

OWL 2 profile validation

FASTO was checked against the OWL 2 DL, EL, RL, and QL profiles. Each profile report was written to a separate text file.

```
for PROFILE in DL EL RL QL; do
  "$ROBOT" validate-profile \
    --input "$FASTO" \
    --profile "$PROFILE" \
    --output "$RESULTS_DIR/FASTO-profile-${PROFILE}.txt"
done
```

ELK reasoning

The ontology was processed with ELK while restricting equivalent-class handling to asserted axioms and excluding duplicate axioms.

```
"$ROBOT" reason \
  --input "$FASTO" \
  --reasoner ELK \
  --equivalent-classes-allowed asserted-only \
  --exclude-duplicate-axioms true \
  --output "$RESULTS_DIR/FASTO-ELK-check.owl" \
  2>&1 | tee "$RESULTS_DIR/FASTO-ELK-reasoner.log"
```

SPARQL inspection queries

The following query families were used to inspect FASTO. Full query text is provided in Appendix B; generated outputs should be retained verbatim.

Query purpose	Evidence produced	Output file
Namespace inventory	Counts declared entities by IRI namespace.	FASTO-namespaces.tsv
Class namespace inventory	Counts declared classes by IRI namespace.	FASTO-class-namespaces.tsv
External entity inventory	Lists declared entities not beginning with the FASTO namespace.	FASTO-external-entities.tsv
External class annotations	Lists labels and definitions for external classes; exploratory only.	FASTO-external-classes-with-annotations.tsv

Query purpose	Evidence produced	Output file
OWL imports	Lists owl:imports statements. The reported FASTO result contained none.	FASTO-imports.tsv
Ontology metadata	Lists properties asserted on the FASTO ontology IRI.	FASTO-ontology-metadata.tsv
Provenance metadata	Finds common provenance and source predicates outside the ontology header.	FASTO-provenance-metadata.tsv
Version statements	Finds versionInfo, versionIRI, priorVersion, and compatibility statements.	FASTO-version-statements.tsv
Declared entities	Lists classes, properties, annotation properties, and named individuals.	FASTO-entities.tsv
DMTO/FTDM namespace subset	Extracts FASTO entities whose IRIs begin with DMTO or FTDM namespaces.	FASTO-DMTO-FTDM-entities.tsv

Query execution pattern

```
"$ROBOT" query \
--input "$FASTO" \
--query "$QUERY_DIR/list-namespaces.sparql" \
"$RESULTS_DIR/FASTO-namespaces.tsv"
```

```
"$ROBOT" query \
--input "$FASTO" \
--query "$QUERY_DIR/list-class-namespaces.sparql" \
"$RESULTS_DIR/FASTO-class-namespaces.tsv"
```

Source-ontology comparison method

```
SOURCE_NAME="SOURCE"
SOURCE_FILE="$REPO_ROOT/data/sources/SOURCE.owl"
```

```
"$ROBOT" query --input "$SOURCE_FILE" \
--query "$QUERY_DIR/list-ontology-declarations.sparql" \
"$RESULTS_DIR/${SOURCE_NAME}-ontology-declarations.tsv"
```

```
"$ROBOT" query --input "$SOURCE_FILE" \
--query "$QUERY_DIR/list-any-ontology-metadata.sparql" \
"$RESULTS_DIR/${SOURCE_NAME}-metadata.tsv"
```

```
"$ROBOT" query --input "$SOURCE_FILE" \
--query "$QUERY_DIR/list-version-statements.sparql" \
"$RESULTS_DIR/${SOURCE_NAME}-version-statements.tsv"
```

```
"$ROBOT" query --input "$SOURCE_FILE" \
--query "$QUERY_DIR/list-namespaces.sparql" \
"$RESULTS_DIR/${SOURCE_NAME}-namespaces.tsv"
```

```
"$ROBOT" query --input "$SOURCE_FILE" \
```

```

--query "$QUERY_DIR/list-entities.sparql" \
"$RESULTS_DIR/${SOURCE_NAME}-entities.tsv"

tail -n +2 "$RESULTS_DIR/${SOURCE_NAME}-entities.tsv" | sort >
"$RESULTS_DIR/${SOURCE_NAME}-entities.sorted.tsv"

tail -n +2 "$RESULTS_DIR/FASTO-${SOURCE_NAME}-entities.tsv" | sort > "$RESULTS_DIR/FASTO-
${SOURCE_NAME}-entities.sorted.tsv"

comm -13 "$RESULTS_DIR/${SOURCE_NAME}-entities.sorted.tsv" "$RESULTS_DIR/FASTO-
${SOURCE_NAME}-entities.sorted.tsv" > "$RESULTS_DIR/FASTO-only-${SOURCE_NAME}-
entities.tsv"

awk -F ' ' '{count[$2]++} END {for (type in count) print count[type], type}' "$RESULTS_DIR/FASTO-only-
${SOURCE_NAME}-entities.tsv" | sort -nr > "$RESULTS_DIR/FASTO-only-${SOURCE_NAME}-
counts.txt"

```

SPARQL query catalogue

B.1 List ontology declarations

PREFIX owl: <http://www.w3.org/2002/07/owl#>

```

SELECT DISTINCT ?ontology
WHERE {
  ?ontology a owl:Ontology .
}
ORDER BY ?ontology

```

B.2 List generic ontology metadata

PREFIX owl: <http://www.w3.org/2002/07/owl#>

```

SELECT ?ontology ?property ?value
WHERE {
  ?ontology a owl:Ontology ;
    ?property ?value .
}
ORDER BY ?ontology ?property ?value

```

B.3 List version statements

PREFIX owl: <http://www.w3.org/2002/07/owl#>

```

SELECT DISTINCT ?subject ?property ?value
WHERE {
  ?subject ?property ?value .
  VALUES ?property {
    owl:versionInfo
    owl:versionIRI

```

```

    owl:priorVersion
    owl:backwardCompatibleWith
    owl:incompatibleWith
  }
}
ORDER BY ?subject ?property ?value

```

B.4 List all declared entities

```

PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

```

```

SELECT DISTINCT ?entity ?type
WHERE {
  VALUES ?type {
    owl:Class
    owl:ObjectProperty
    owl:DatatypeProperty
    owl:AnnotationProperty
    owl:NamedIndividual
  }
  ?entity rdf:type ?type .
  FILTER(isIRI(?entity))
}
ORDER BY ?entity ?type

```

B.5 List namespaces used by declared entities

```

PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>

```

```

SELECT ?namespace (COUNT(DISTINCT ?entity) AS ?entityCount)
WHERE {
  VALUES ?type {
    owl:Class
    owl:ObjectProperty
    owl:DatatypeProperty
    owl:AnnotationProperty
    owl:NamedIndividual
  }
  ?entity rdf:type ?type .
  FILTER(isIRI(?entity))
  BIND(REPLACE(STR(?entity), "^(.*[/#])[/#]*$", "$1") AS ?namespace)
}
GROUP BY ?namespace
ORDER BY DESC(?entityCount)

```

B.6 List class namespaces

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
SELECT ?namespace (COUNT(DISTINCT ?class) AS ?classCount)
WHERE {
  ?class rdf:type owl:Class .
  FILTER(isIRI(?class))
  BIND(REPLACE(STR(?class), "^(.*[/#])[/#]*$", "$1") AS ?namespace)
}
GROUP BY ?namespace
ORDER BY DESC(?classCount)
```

B.7 List external entities

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

```
SELECT ?entity ?type
WHERE {
  VALUES ?type {
    owl:Class
    owl:ObjectProperty
    owl:DatatypeProperty
    owl:AnnotationProperty
  }
  ?entity rdf:type ?type .
  FILTER(isIRI(?entity))
  FILTER(!STRSTARTS(STR(?entity),
    "https://bioportal.bioontology.org/ontologies/FASTO/"))
}
ORDER BY ?type ?entity
```

B.8 External classes with labels or definitions

```
PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX iao: <http://purl.obolibrary.org/obo/IAO_0000115>
```

```
SELECT DISTINCT ?class ?label ?definition
WHERE {
  ?class rdf:type owl:Class .
  FILTER(isIRI(?class))
  FILTER(!STRSTARTS(STR(?class),
    "https://bioportal.bioontology.org/ontologies/FASTO/"))
  OPTIONAL { ?class rdfs:label ?label }
  OPTIONAL { ?class iao:IAO_0000115 ?definition }
  FILTER(BOUND(?label) || BOUND(?definition))
}
ORDER BY ?class
```

B.9 List owl:imports

PREFIX owl: <http://www.w3.org/2002/07/owl#>

```
SELECT DISTINCT ?ontology ?importedOntology
WHERE {
  ?ontology a owl:Ontology ;
            owl:imports ?importedOntology .
}
ORDER BY ?importedOntology
```

B.10 FASTO ontology metadata

PREFIX owl: <http://www.w3.org/2002/07/owl#>

```
SELECT ?property ?value
WHERE {
  <https://bioportal.bioontology.org/ontologies/FASTO.owl>
    ?property ?value .
}
ORDER BY ?property ?value
```

B.11 Provenance metadata

PREFIX owl: <http://www.w3.org/2002/07/owl#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dc: <http://purl.org/dc/elements/1.1/>
PREFIX dct: <http://purl.org/dc/terms/>

```
SELECT DISTINCT ?subject ?property ?value
WHERE {
  ?subject ?property ?value .
  VALUES ?property {
    owl:versionInfo
    owl:versionIRI
    rdfs:isDefinedBy
    rdfs:seeAlso
    dc:source
    dct:source
    dct:isVersionOf
    dct:hasVersion
    dct:provenance
  }
  FILTER(?subject !=
    <https://bioportal.bioontology.org/ontologies/FASTO.owl>)
}
ORDER BY ?subject ?property ?value
```